

Vyavastha – A 5G-Powered V2X Solution for Indian Roads

Submitted by Team Pragati Pravaha

The Problem

India has some of the most dangerous roads in the world, with over 150,000 fatalities annually due to traffic accidents. Urban centers like Delhi suffer from unmanaged traffic, frequent congestion, and a lack of coordination among vehicles. Meanwhile, rural highways face low visibility, unexpected hazards like cattle, and limited emergency services. The common challenge is a lack of real-time, connected, and intelligent systems to prevent accidents and enable responsive mobility.



Our Solution – Vyavastha

Vyavastha is an intelligent, scalable Vehicle-to-Everything (V2X) communication framework built for India's roads. Using 5G, Edge AI, and C-V2X technology, Vyavastha creates a digital ecosystem where vehicles communicate in real time with each other, traffic infrastructure, pedestrians, and cloud services. It enables predictive crash prevention, optimized traffic flow, emergency routing, and real-time hazard detection.

Core Features

- **Vehicle-to-Vehicle (V2V):** Alerts drivers of vehicles in blind spots, sudden braking ahead, or lane merging—even around corners.
- **Vehicle-to-Infrastructure (V2I/I2V):** Smart traffic signals adapt to real-time traffic density and prioritize ambulances.
- **Vehicle-to-Network (V2N):** Live data shared with cloud services for traffic prediction and route recommendations.
- **Vehicle-to-Pedestrian (V2P):** Smartphones alert both drivers and pedestrians about crossing movements or risks.

System Architecture

Vyavastha is built on a modular and future-ready architecture that ensures real-time, intelligent communication across every layer of the transportation ecosystem. The system comprises the following core components:

- **On-Board Units (OBUs):**

Installed in vehicles, OBUs continuously collect and broadcast telemetry data such as speed, location, direction, and sudden braking events. These units enable **Vehicle-to-Vehicle (V2V)** and **Vehicle-to-Infrastructure (V2I)** communication in real-time.

- **Roadside Units (RSUs):**

Strategically placed at traffic junctions, highway checkpoints, and pedestrian crossings, RSUs receive incoming data from nearby vehicles and respond with context-aware information—like hazard alerts or dynamic signal control. They act as local hubs for **Vehicle-to-Infrastructure (V2I)** and **Infrastructure-to-Vehicle (I2V)** flows.

- **5G Core Network:**

Vyavastha leverages India's emerging 5G infrastructure to enable ultra-reliable, low-latency communication. Using **URLLC (Ultra-Reliable Low-Latency Communication)**, safety-critical messages like crash alerts and braking signals are delivered in milliseconds. **eMBB (Enhanced Mobile Broadband)** supports high-bandwidth data like map streaming, dashcam footage, and traffic analytics.

- **Multi-access Edge Computing (MEC):**

Edge servers are placed closer to RSUs to offload decision-making from the cloud. These servers run **AI algorithms** that predict potential collisions, manage signal coordination, and reroute vehicles in emergencies. Edge computing ensures decisions are taken within milliseconds, even in network-congested zones.

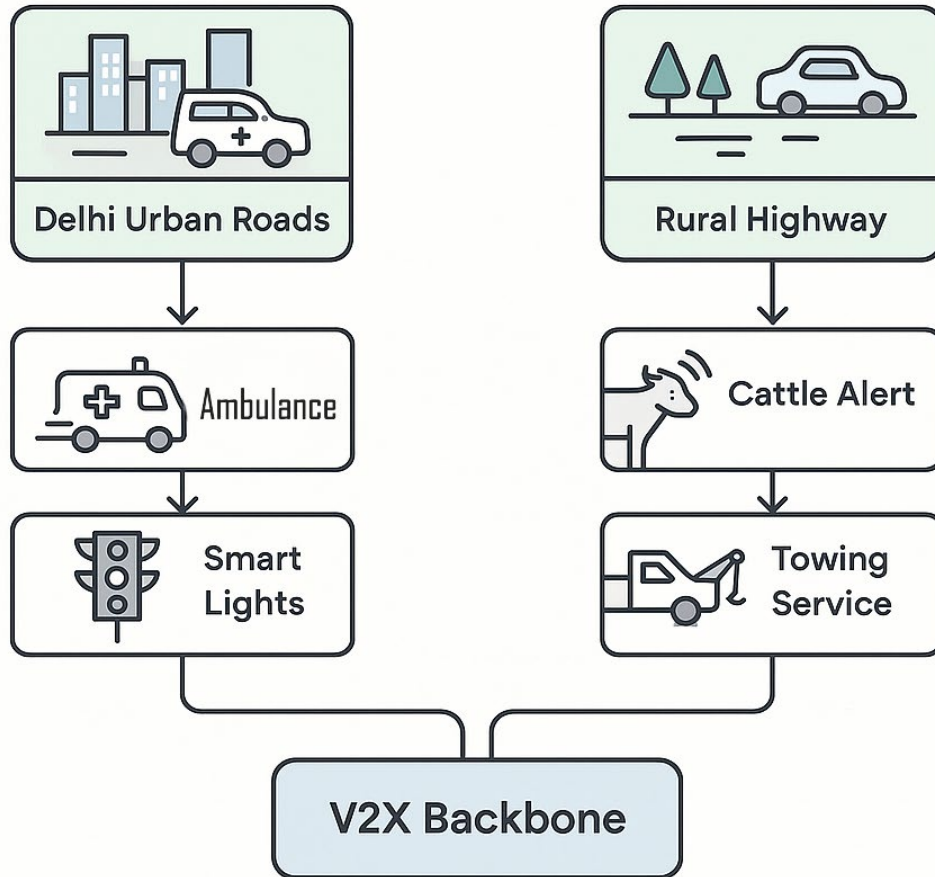
- **Cloud Platform & Backend Analytics:**

While edge nodes handle local decisions, the cloud platform performs long-term analysis and coordination. It provides real-time dashboards to city authorities, supports emergency service dispatching, and helps urban planners redesign road layouts using collected V2X data.

- **Data Flow & Coordination:**

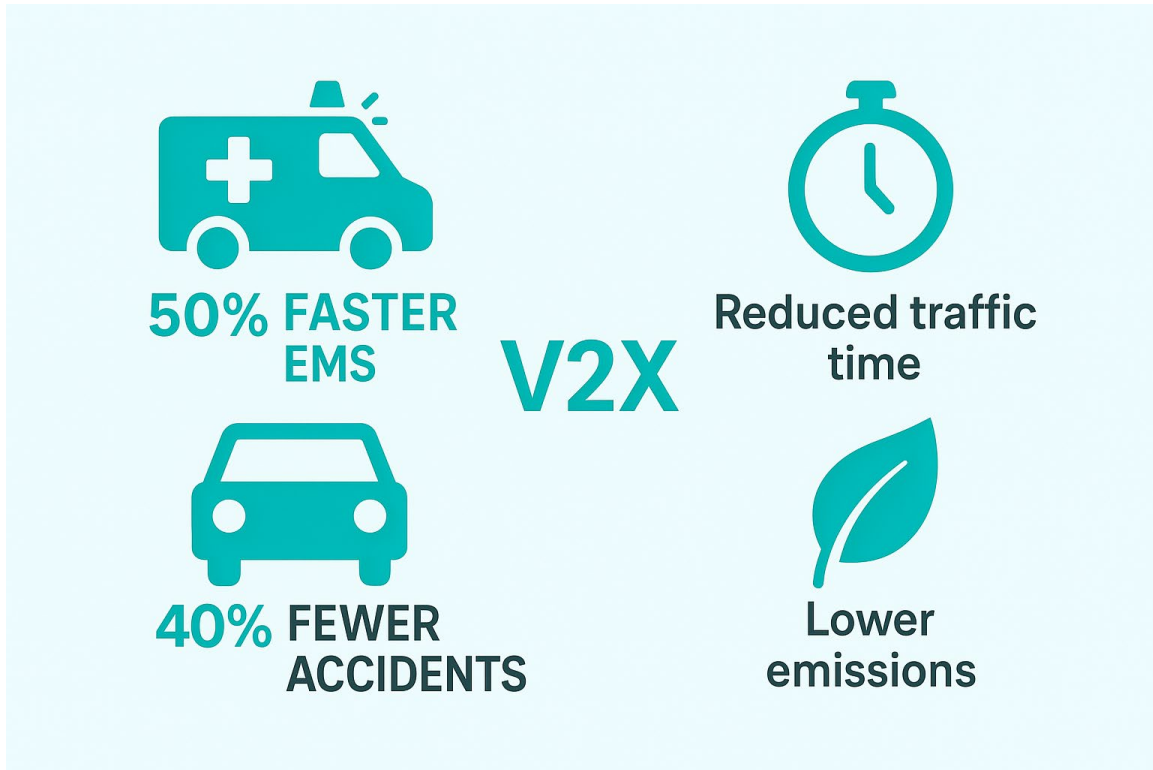
All data flows—V2V, V2I, V2N, V2P—are synchronized to form a **dynamic, self-healing mesh network**. Each node (vehicle, infrastructure, or server) contributes to safer and more efficient road mobility.

Deployment Flowchart (Urban + Rural)



Impact

- Up to 40% reduction in accidents through proactive warnings.
- 50% faster ambulance response using smart traffic routing.
- Improved traffic fluidity, fuel savings, and reduced pollution.
- Real-time data for smart city planning and infrastructure improvements.



Prototype Plan (Hackathon Phase)

To validate the Vyavastha system during the hackathon, we will develop a **scaled-down working prototype** that simulates key functionalities of V2X communication, edge processing, and real-time visualization.

1. V2V & V2I Communication Simulation

- Raspberry Pi and Arduino boards will emulate **On-Board Units (OBUs)** and **Roadside Units (RSUs)**.
- Vehicles will broadcast simulated telemetry (speed, location, direction) to each other and to RSUs.
- RSUs will control virtual traffic signals and respond to high-priority messages (e.g. ambulance routing).
- V2V alerts like **sudden braking or blind-spot detection** will trigger dashboard warnings on receiving nodes.

2. Edge Server with AI Collision Prediction

- A **local edge node** (laptop or Jetson Nano) will run lightweight AI models.
- Based on vehicle trajectories, it will predict potential collisions and issue warnings in milliseconds.
- Demonstrates **real-time decision-making without cloud latency**.

3. Real-Time Visualization Dashboard

- A **web-based live dashboard** (built with HTML5, JS, and Google Maps API) will show:
 - Connected vehicles
 - Route status
 - Hazard locations
 - Suggested diversions
- Live MQTT feeds will update the dashboard instantly.

4. Communication Stack: Python + MQTT

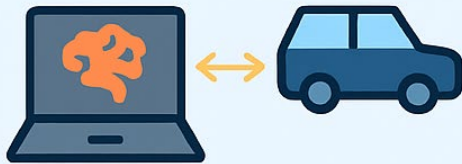
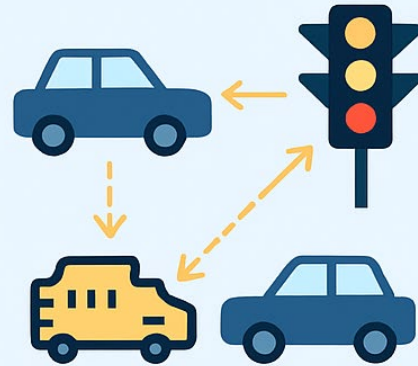
- Devices will use **Python** and **MQTT protocol** for reliable low-latency messaging.
- Custom scripts will handle telemetry data exchange, event detection, and UI rendering.

5. Enhancements

- Integration with **5G dongle or cellular fallback** (for V2N simulation).
- Basic V2P alert from a pedestrian's smartphone using a web-app beacon.
- Use of **YOLO object detection** for RSU-level hazard detection via camera.

1. V2V & V2I Communication Simulation

- Raspberry Pi and Arduino devices emulate V2V-Alerting and V2I
- Broadcast simulated telemetry to each other and to VTS
- V2V alerts*virtual traffic signals and respond to high-priority messages
- Alerts like sudden braking or blind-spot detection



2. Edge Server with AI Collision Prediction

- A local edge node running lightweight AI models
- Predict potential collisions and issue warnings in real-time decision-making without cloud latency

3. Real-Time Visualization Dashboard

- A web-based live dashboard (built with HTML5, JS, Google Maps API)



Connected vehicles, Route status, Hazard locations and suggested diversions

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Future Vision

Vyavastha is designed to scale across Indian cities and highways, integrating with Smart City programs, supporting autonomous vehicles, and making our roads safer and more intelligent. It aims to become India's national V2X layer for next-generation mobility.

Team Details

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